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ADNOC Offshore Digital Factory: Advanced Decision-Making via System Performance and Risk Quantification

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Abstract

Leveraging the innovative ADNOC Offshore Digital Factory's concept of Reality Based Prescriptive Predictive Maintenance (RB- PsM/PdM) High-Fidelity models to predict future availability and throughput for each production stream, as well as the revenue and costs, and to identify those units which are the main contributors to loss of performance (culpabilities), in addition to generate additional 18+ Key Performance Indication represents the efficiency and reliability for the production stream as well as individual areas/sites.

This approach aims to evaluate the impact of changes at the equipment, operations, and maintenance levels to enhance availability and throughput, assesses strategic and operational scenarios based on cost-benefit analysis, hence provides a digital solution for supporting business decision-making processes.

ADNOC Business Outcomes:

- Informed Standard Decision-Making Process: quantify the benefits and costs of options to modify asset operation & maintenance strategy or system design, as well as the probability of achieving an outcome.
- Improve Reliability: identify and address the elements in the system that contribute the most to non-performance and the root causes of downtime, leading to significant improvements in reliability and performance.
- Improve Availability: assists in improving the availability of assets, units, and systems, supporting assessment of changes to operating and maintenance strategies, spare's philosophy, design and/or logistics.
- Elevate Predictability: enhances the ability to predict and foresee system behavior, helping you better prepare for potential scenarios and outcomes.
- Decrease Life Cycle Cost: help in reducing the overall life cycle costs associated with assets and systems by identifying areas for optimization and cost reduction.
- Optimized Operations Strategy: helps in streamlining operations strategies for optimum performance and maximum profitability.
- Improve Maintenance Strategy: enhance maintenance strategies and procedures for prolonged asset life and minimal downtime.

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- Optimize CapEx and OpEx: enable quantification and comparison of the benefits and costs of different alternatives.

Introduction

ADNOC Offshore (AOF) Digital Factory Project Initiative is an AI/MP based Hub mandate is to deliver Top-side Monitoring & Decision-Making Center.

The whole concept is based on AOF UZ's new innovative concept of Integrated Total Reliability Solution (TRS) with Extended Accuracy patented Reliability Based Prescriptive Predictive Maintenance (RB-PsM/PdM) Platform, provided by a consortium of multiple technology providers.

The MLP will enable AOF to offer a truly unique concept for equipment reliability in the oil & gas industry. The MLP will contribute to reducing operating cost by optimizing IMR (inspection, maintenance, and repair) activities, deferring unnecessary future capital investment through novel asset lifetime extension algorithms, improving performance while maintaining asset integrity, enhancing safety, and minimizing production loss.

It's based in "Defense In Depth" (DID) principle to provide a remote real-time Online/ Offline advanced Proactive Predictive Monitoring for High, Medium and Low Critical Top-side Machinery/ Assets (Rotating, Static Electrical and Control Equipment/ assets).

ADNOC Digital Factory supports operations in following ways:

- Maximize the uptime
- Reduce Infant mortality period & increase wear-out period, etc.
- Increase MTBO (Mean Time Between Overhaul) / MTBI (Mean Time Between Inspection) period • Prolong / Extended the usual lifetime cycle of the Equipment, Systems and Plants (Increase Constant failure rate period)
- Increases maintenance efficiency and eventually maximize Plant / gross assets uptime
- Increase Production Availability / Efficiency by avoiding unplanned production deferrals and costly breakdown
- In-depth design step of unlock Potentials and Debottleneck Facility Limitations to improve efficiency and effectives for plant/ propel (OPEX) and future projects (CAPEX)

The project concept starting point is to master the downtime events and match them to probable causes that are trapped in downstream data silos:

- To identify the "Bad Actors" through Bad Actors Management System, (BAMS) and potential root causes and ordering them by equipment that is causing the most production loss
- Employ Artificial Intelligence & Machine Learning to automatically identify early warnings of defects and even before defects start
- Center of Expertise to monitor the trends, review alerts and provide actionable advisory to all assets
- Continuously improve the Expert System leveraging the ADNOC economy of scale.
- Standardization, internal benchmarking, lessons sharing
- Center of Excellence for Machinery and Asset Management Expertise to develop and maintain competency standards in all ADNOC and beyond
- Masters' downtime events and clusters them by equipment, and finally Matches these events to Notifications that contain technician notes that describe probable root cause and fixes, Identify productivity loss actors.

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The Machinery Lab platform envelope covers the following AOF UZ's in house narrated procedures/ Code Of Practice "COP" as well as international standards/ COPs:

- Condition Monitoring "CM"/ Maintenance Management System "MMS"/ Condition Based Maintenance "CBM"/ Reliability Based Predictive Maintenance (RB-PdM®) using Wired/ wireless assets connectivity
- Root Cause Analysis "RCA"/ Asset Management System "AMS"/ Asset Integrity Management System "AIMS"/ Bad Actors Management system "BAMS"
- Continue and extended using ADNOC's Computerized Predictions Analytics and Diagnostics "CPAD" Modeling/ Perdition platform (by Honeywell and others)
- Adopt new Perdition concept of DIGITAL TWINS & RB-PdM innovation Plant Protection System
- Online/ Offline Using Adnoc Offshore Condition Monitoring system (all Protocols)
- Process Simulation (DESIGN Boundaries/ Process Eng.)
- MAINTAIN/ Driving uptime with APM
- RAM / ROM & LCCA Modeling
- Augmented Reality "AR"/ Artificial Intelligence "AI" Digital Outages "DO" / smart maintenance system
- Reduced Based-FEA / CFD Modeling
- OPERATE/ Running to the limits of performance
- Asset Life Cycle (e.g. Asset Replacement Plan "ARP")
- Manufacturing & Supply Chain
- R&D (Debottleneck Facility Limitations)
- Full bidirectional Connectivity with CMMS/SAP
- Full Connectivity with Operator Driven Reliability Platform "ODR"
- Multilevel KPIs

AOF aspires to combine novel technologies such as data-driven modelling using artificial intelligence/ machine learning techniques (AI/ML) and physics-based advanced Reality-Based Predictive/ Prescriptive maintenance (RB-PsM/PdM®) with Condition-based Monitoring (CBM) technologies using hybrid analytics in the Asset Performance Management platform. Thus, the MLP shall leverage cutting edge solutions to achieve the maximum possible levels of accuracy to predict equipment failure and thus ensure sustainable and cost-effective production.

AOF is currently deploying Asset Monitor and Predictive Analytics, branded as CPAD (Centralized Predictive Analytics and Diagnostics) program. CPAD underpins ADNOC's 2030 smart growth strategy and Oil & Gas 4.0 initiatives, aims at transforming the company's operations to maximize value from every barrel of oil, while delivering the greatest possible returns to the UAE.

The Digital Factory will be a truly novel technology-led approach to asset integrity offering a combination of condition monitoring, integrated & contextualized data management and physics-based and ML-driven models to:

1. Monitor in near-real time the true structural health of the asset,
2. Predict future health condition with one of the highest accuracy predictions,
3. Extend the asset lifetime better than any other tool on the market,

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4. Provide an integrated APM (Asset Performance Management) platform to enable the use cases and work practices to:
   a. Assure continuous improvement workflows that future proof the solution,
   b. Enable diagnostics, troubleshooting and integrated root cause analysis of equipment failures, leveraging static and dynamic data and modelling algorithms as part of the APM ecosystem,
   c. Improve and update asset strategy management and maintenance strategies,
   d. Provide a unified view of asset reliability health and condition risk scores to measure cost, output, failure rates and compliance metrics,
   e. Increase machine reliability and availability and reduce maintenance cost by minimizing unplanned or forced outages and by ensuring maintenance strategies are current and optimized,
   f. Increase safety and reduce potential environment impacts,
   g. Offer a single source of asset health intelligence including advanced visualization and analytics of asset health, asset strategy and work execution data
   h. Enable direct connectivity to work execution to accelerate maintenance activity and reduce MTTR (Mean Time to Repair).

To achieve these goals, the proposed technologies must be the best of breed in their respective domains to support ADNOC deliver on its 100x acceleration initiative, focus on automation and remote monitoring, reduction/ elimination in asset redundancy and lower cost per barrel. The key components of the Digital will be:

- Engineering simulation platform - to build physics-based sensor-enabled structural digital twins to monitor the structural health of critical assets in near-real time. The technology must be 1000x faster, unlimited by scale and at least 10x more accurate than existing FEA (Finite Element Analysis) tools available in the market currently.
- Fully integrated APM - includes asset health management, asset strategy management, and defect elimination, enabled by analytic insights, build on Bently Nevada System 1's best-in-class plantwide condition monitoring capabilities and covering the entire enterprise.
- Process simulation - industry leader, comprehensive, intuitive, and enables optimization by AI
- Multivariate analysis - to rapidly identify the process parameters most critical to quality, cycle time and other key KPIs for production support
- System Performance, Risk and Reliability (RAM/ROM) simulation - to maximize the economics of business decisions well beyond the equipment level and accurately predict future asset performance of the whole system
- Data integration & contextualization - to make asset data more accessible and meaningful, trusted, and reliable data to produce insights that unlock opportunities in real-time.
- Bad Actors Management (BAMS) direct connectivity to work execution to accelerate maintenance activity and reduce MTTR (Mean Time to Repair).

Methodology

The System Performance, Risk and Reliability (RAM/ROM) simulation methodology relies on a high-fidelity integrated model for operation facilities of ADNOC Upper Zakum-Zirku stream (which was extended later to Lower Zakum & SARB streams), including wells, gas-lift and injection units, separation units, compression systems, pipelines, storage elements, operational logic that accounts for separation splits and demurrage, well production profiles, demand profile, and equipment reliability, thus capturing the variability and uncertainty in the system over time.

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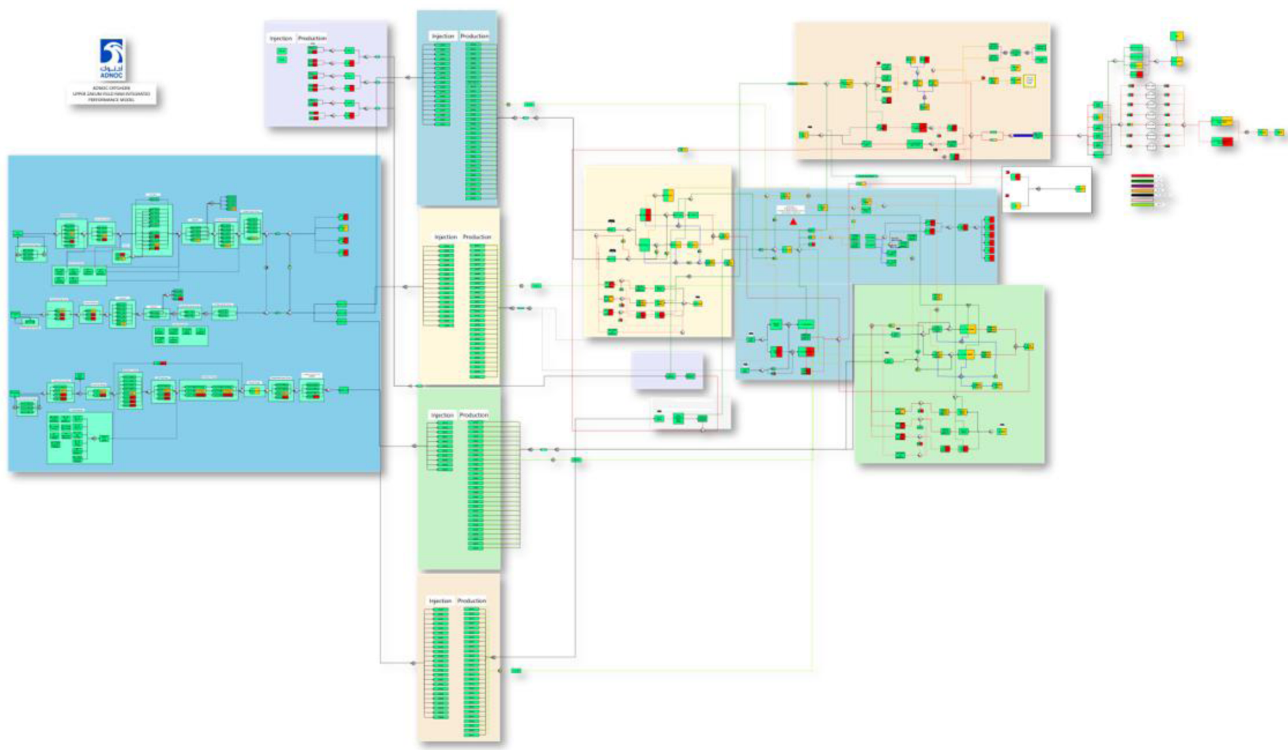
The model is used to generate assets availability and production throughput forecasts via advanced multi-state discrete-event simulations using Monte-Carlo with flow optimization and includes the execution of custom logic incorporated by the user. utilizing statistical sampling techniques plus planned schedules in order to create stochastic and deterministic events that impact the system and solves the network system diagram and then predicts the future performance of the system.

The model flow diagram is composed of units containing the components/events that have critical impact to the overall performance of the system. As each event occurs, it influences the unit that it is in. In turn, each unit influences the flow sent through the pipes attached to it, both upstream and downstream. The Simulator tracks the flow through the pipes as well as the utilized and available capacities of the units. Note: pipe flow can be product, utility, power, emissions, revenue, costs, etc.



Figure—Model Flow Diagram

The study process begins with an "as-operated" or "as-designed" process flow model of the ADNOC stream. update of the production profiles, demand profile, asset reliability data register (list of equipment/ events and each item's frequency and duration distribution), plant design capacities and operating rules constructed for the model, then the simulation is triggered to generate new forecasts, and based on that identify bottlenecks. Subsequently, followed by What-If Analysis of different asset operation or maintenance strategies or system configurations can be simulated too in order to identify availability and production throughput improvements.

Input data supplied through ADNOC Data Fusion Platform and set to run the simulation periodically and update the enterprise dashboards with the exported KPI Output from the model.

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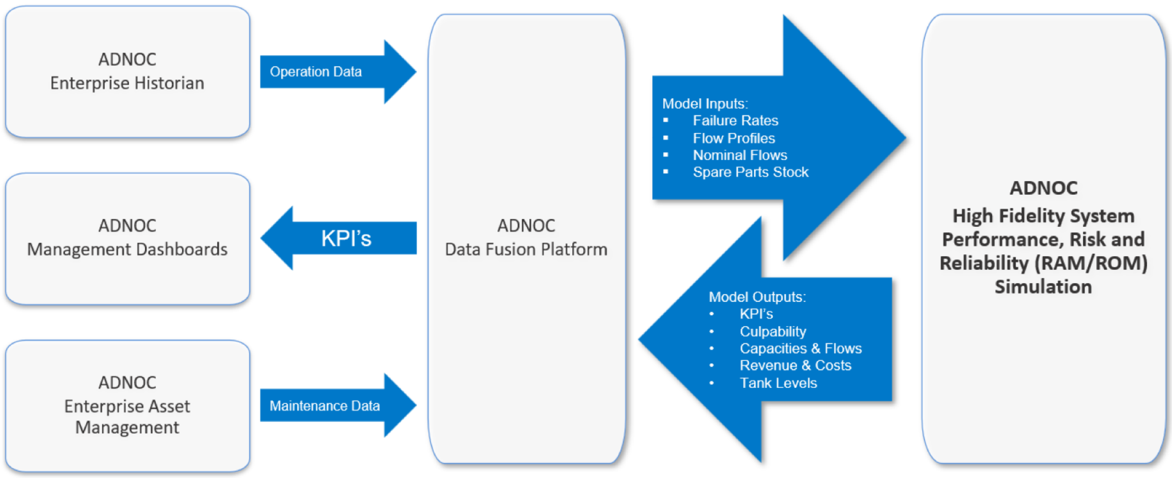

Figure—Sampling Process

Modeler/User typically specify the inputs in Fidelis model, which include the connectivity between processing units (which can represent a single asset or multiple assets) and storage elements, set nominal flows in and out the units, storage capacities, and configure deterministic and stochastic events and their impact to the units (where events can represent mechanical failures of equipment, planned preventive maintenance activities, arrivals or departures of ships, extreme weather events, supplier delays, green field projects construction start times of units, and more), operational rules and logistics, lifecycle duration and number of lifecycles to simulate.

The behavior of equipment in the system represented by planned and unplanned maintenance is represented by a probabilistic distribution function. This function is derived by using curve-fitting techniques on the raw historic data or asset remnant life or upstream industry standard data. The curve that is generated represents the possible behavior of the event (events include equipment failures and repairs, changes in weather, demand, pricing, feed rates, etc.). The solution platform samples from this curve and derives possible time- to-event behavior (frequencies and durations) for each event. The Time and duration of events are sampled along a given period, and at every sampled time; production flow network optimized based on given rules and priorities.

Many lifecycles are run in order to sample enough points from all of the events in the model to ensure that a statistically relevant number of possible combinations are considered. Each lifecycle predicts a possible future of the actual system. After many lifecycles, a histogram of the possible outputs is created. In this way, the results represent the probability or confidence of meeting or exceeding any KPI, not just an average or mean.

Simulation will run number of lifecycles specified by the user. For each lifecycle, start at time zero, and then move to the next event time, until the end time (i.e., the lifecycle duration specified by the user) is reached.

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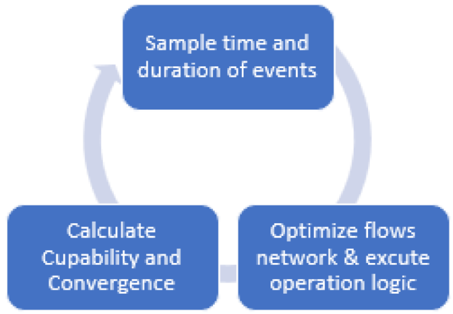

Figure—Model Flow Diagram

1. sample events time and duration along a given period.
2. solves a Linear Optimization model to calculate the flows across the system.
3. executes the additional operation logic coded through Visual Studio 2019 interface VB.NET.
4. calculate Culpability (contributions of events to losses of performance) and convergence values.

After all lifecycles are simulated, Fidelis Simulator will create different charts (histograms, trends, Pareto charts) for the users to visualize the simulation results for the forecasted future system performance, which include unit availability and utilization, production throughput, revenue, emissions, costs, tank levels, spare parts coverage and Culpability to identify top bad actors and assess the value of improvement alternatives.

Structural Elements

Model Flow Diagram are the top level of structural hierarchy for the model. Which is basically a hybrid of block diagrams and process flow diagrams.

Flow Diagram elements placed & connected in a way that represents the real-world system configuration. The elements (building blocks) that comprise flow diagrams are units, pipes, manifolds, and tanks. These elements are used to define a network of ADNOC Offshore assets. The network, or system, has behaviors that are determined by how the constituent parts connected and by an underlying LP optimization engine (optimizer) whose purpose is to optimize throughput of the system at each point in time.

The optimizer controlled by specification of hierarchical goals (Objective Function) which provided by users. These goals are specified in terms of priorities for units' utilization and/or pipe flow rates. The priorities can either be static or can be modified dynamically as a simulation progresses in time. Dynamic prioritization is enabled by use of custom coding provided by users.

flow diagram elements.

a. Units: Units used to group assets to functionally related sets of equipment.
b. Pipes: Pipes used as flow connections among units, manifolds, and tanks.
c. Manifolds: Manifolds serve a modeling purpose to gather and distribute.
d. Tanks: Tanks are high fidelity representations for store bulk material physical analogs.

Unit diagram elements.

a. Event: Represents individual pieces of equipment events, or preventative maintenance schedule or external disturbances.
b. Link: Represents transportation of available capacity information from one unit element to another unit element.
c. Rate Multiplier: Represents a header to create parallel events in the unit diagram.
d. Start and End Node: Represents the source and sink of your unit diagram.

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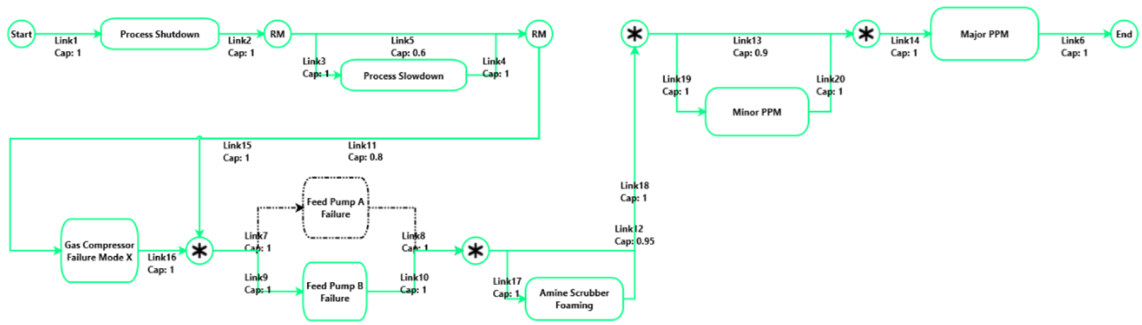

Figure—Unit Diagram General Example

Challenges & Limitations

A common key concern in simulation is the influence of process conditions and operating efficiencies. Factors such as loads, variable feed quality, separation efficiency and process dynamics can significantly impact system performance but are often simplified or excluded in simple Risk models. This introduces additional complexity when seeking realistic production forecasts. However, these limitations can be mitigated by integrating RAM simulations with process models and embedding custom logic code, allowing the model to account for real operating conditions, efficiency variations, and system interactions, thereby improving accuracy and decision-making.

Analysis Results

Aspen Fidelis makes use of the Monte Carlo method to model real-world behavior that are non-deterministic by nature. Statistical distributions are sampled to determine values that modeled attributes take on at discrete points in time. Examples include time between equipment failures, duration of repairs, impacts of failures (on/off or partial operation), preventive maintenance schedule, spare parts availability, human factors, ambient conditions, etc. Random number algorithms provide the mechanism for probabilistic selection of attribute values as simulation time progresses. Built-in optimization with hierarchical goals (static and/or dynamic) is applied to the flow network at each point in time to maximize throughput.

This approach quantifies time-dependent system (and sub-system) behavior, both as mean values and as distributions. Statistical validity (convergence) is achieved by multiple, independent trials, or "lifecycles" in Fidelis terminology, each of which applies to a time interval of interest called "mission time". "Mission time" for the simulation defines a period of interest over which behavior is modeled, typically multiple years for Oil & Gas plants and similar industrial assets.

Outcomes from each lifecycle are tracked and quantify both averages and variability of outcomes. This provides extra dimensions of information and insight: quantification of downside risk and upside opportunity.

The simulation results includes 18+ different production KPI's including the predicted availability and utilization of the different equipment, and throughput for each production area in the model, as well as a ranked list of equipment based on their contribution to production losses. The base case for this model has been validated by subject matter experts, and some scenarios have already been simulated to quantify the business impact of equipment shutdowns and terminal unavailability on the overall stream. The value of these results has led to increasing the scope of the model.

Capacity.

- Available Capacity - This is the ability of the unit to produce based on both the mechanical and operational state of events comprising the model. This gives you the availability of each modelled unit as if the unit was disconnected from the remainder of the system.

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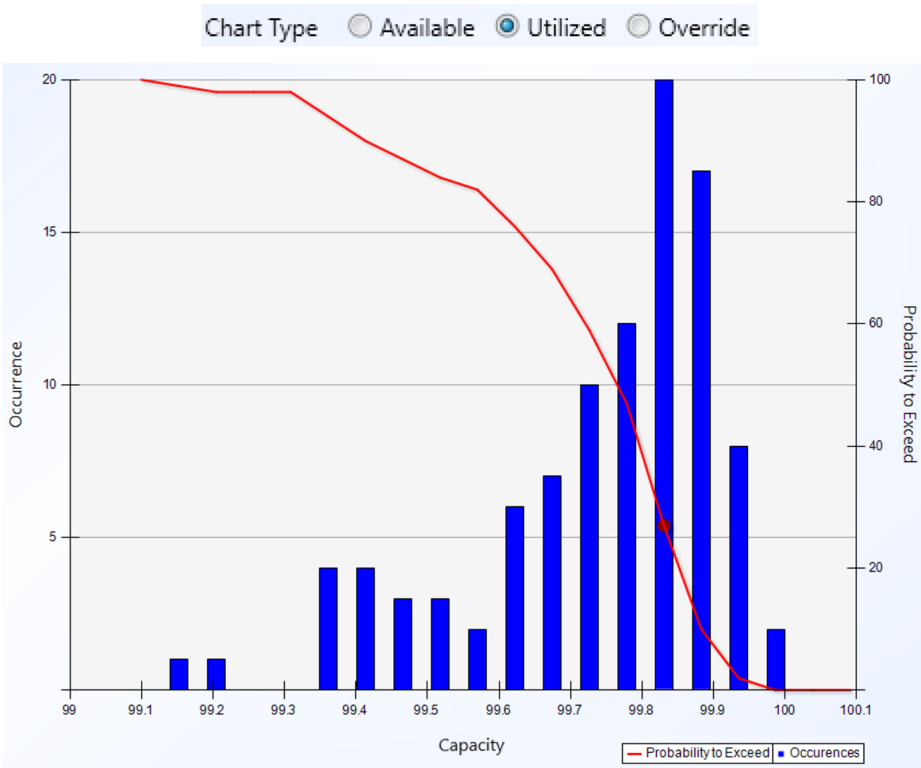
- Utilized Capacity - This is the actual production of the unit based on all factors including equipment, active status, operational status, external influences of other units and tanks, and user manipulation of override capacity. If there is no turn-up capacity in the model, then utilized capacity is equal to or less than the available capacity.
- Override Capacity - This is a property that you can use to check if turn-up capacities influence unit capacities. Override capacity takes precedence over available capacity and the only factors potentially constraining a unit below this value are constraints from other units and tanks. The override capacity values are the same as the available capacity values unless altered in the Write Key Routines.



Figure—Stream Overall Capacity Histogram Chart

Unit Capacity - Quantifies the Available and Utilized Capacities of each unit and for the entire system to find bottlenecks, shows average capacity across entire lifecycles.

Unit Capacity Over Time - Identifies the variability in unit usage over the entire time sequence, highlights seasonal and other time- based variability, shows how capacity changes over time within lifecycles.

Unit Convergence - Shows the average unit usage after each lifecycle has been simulated, helps to identify if more lifecycles need to be simulated if desired convergence is not achieved, shows how the confidence in capacity estimate changed as more lifecycles were run.

Shutdowns and Slowdowns - Insight into behavior of each unit for internal and external issues, also helps with model validation, shows how many times events did not have 100% capacity.

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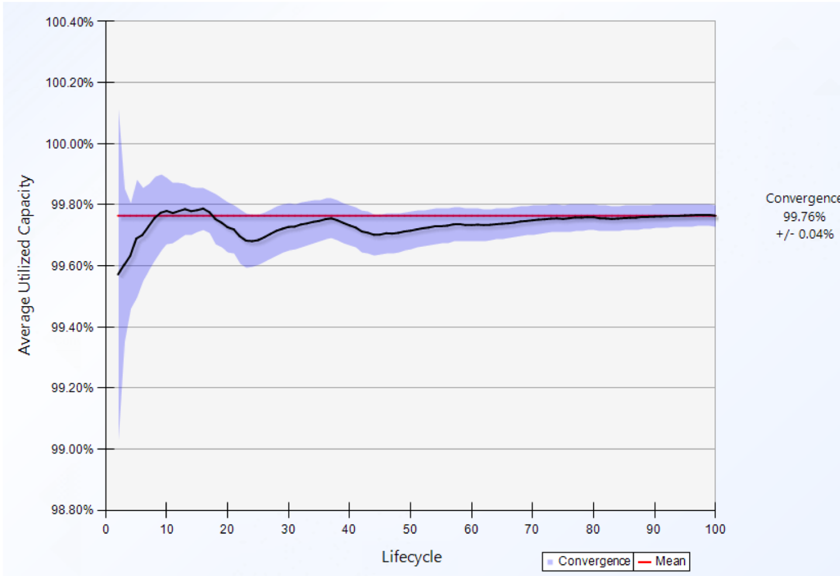
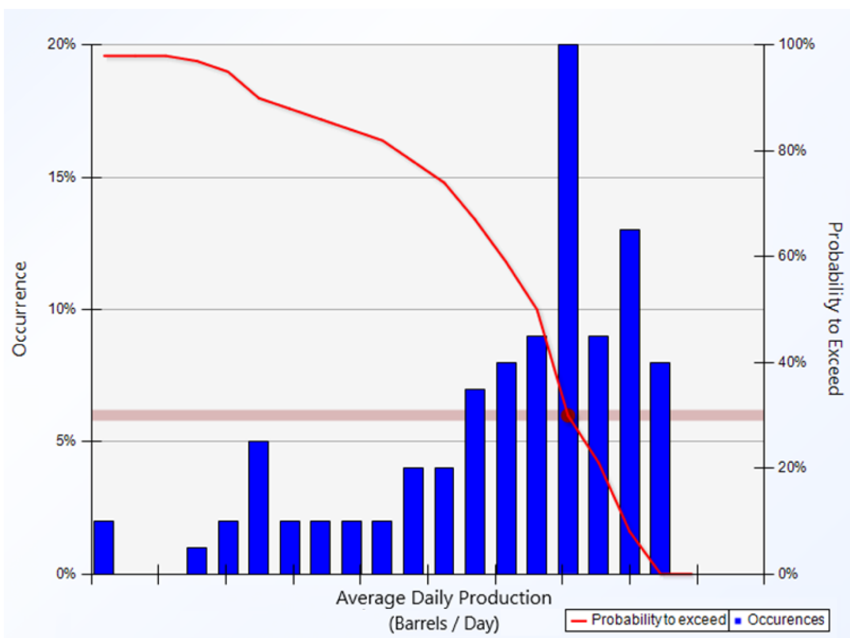


Figure—Stream Overall Convergence Chart

Flow. Production Flow: Quantifies cumulative flow for model simulation time through any chosen pipe (production, revenue, emissions, etc.), also shows average value per hour, day, week, month, and year.

Production Flows: Shows flow vs time for any selected pipe, identifies any time-based variability.



Figure—Stream Overall Production Histogram Chart

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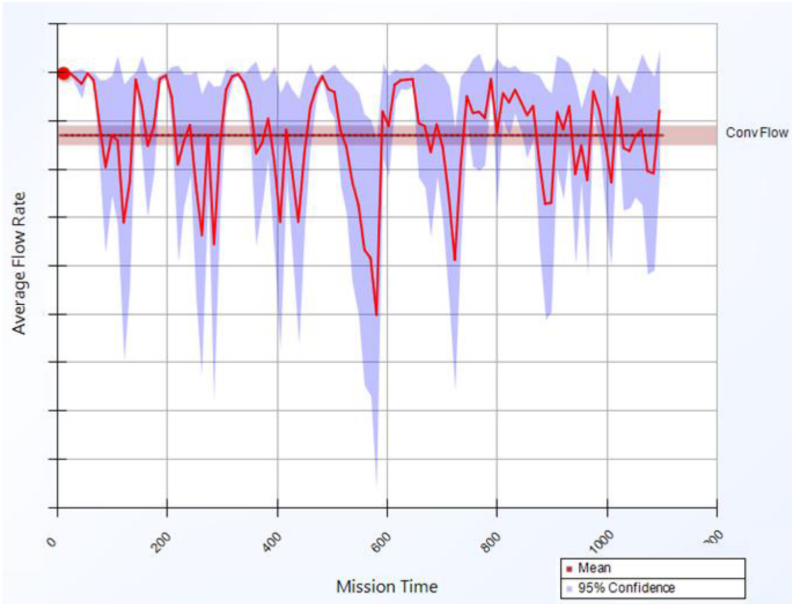
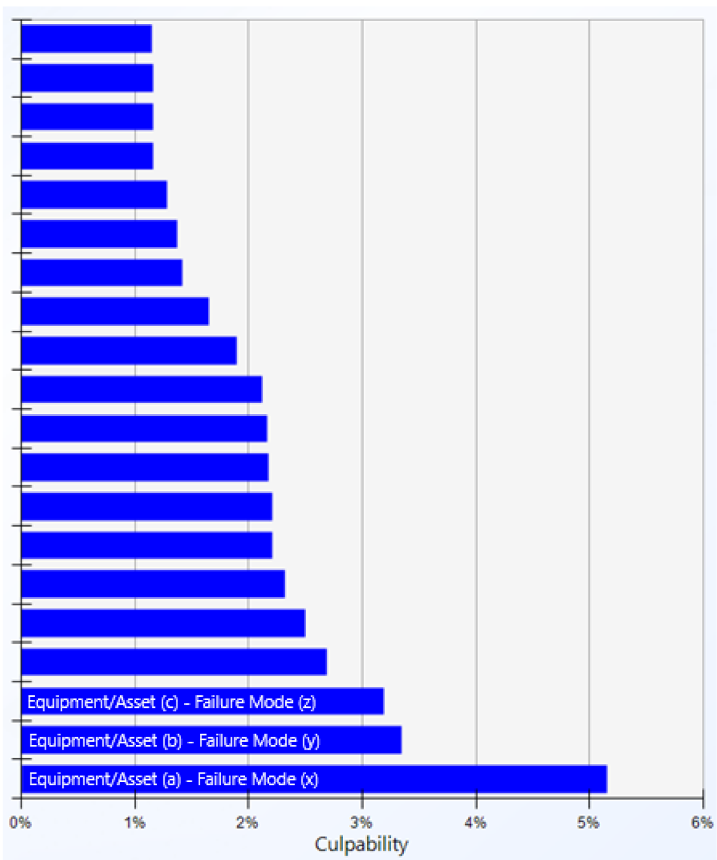


Figure—Stream Overall Production Flow Trend

Culpability. Main Contributors to loss of Performance: Quantifies the impact that each equipment or event has on KPI's (production, revenue, availability, emissions, etc.), identifies top bad "actors" on which to focus improvement opportunities and investment.



Figure—Stream Culpability Chart

Culpability values are calculated with respect to the selected Total Production Basis Unit.

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Key Performance Indicators. Additional Custom KPI's: 18+ different production KPI's including the predicted availability and utilization of the different equipment, and throughput for each production area in the model

Table—Additional KPI's & Calculations

#	Description	Units	Calculation
1	Actual Production Efficiency	%	*[Actual Production / Design Production Capacity]*
2	Production Availability	%	*[Actual Production / (Actual Deferral + Actual Production)]*
3	Production Reliability	%	*[Actual Production / (UnplannedDeferral + Actual Production)]*
4	Total Production Efficiency	%	*[Actual Production /Full Installed Capacity]*
5	Operational Production Efficiency	%	*[Actual Production /Required Production]*
6	Operational Availability	%	*[Up Time / Total Time]*
7	Potential Efficiency	%	*[Required Production / Design Production Capacity]*
8	Required Efficiency	%	*[Required Production /Potential Production]*
9	Average Efficiency	%	*[Predicted Achieved Production /Potential Production]*
10	Predicted Production Efficiency	%	*[Predicted Achieved Production /Required Production]*
11	Offshore Production Efficiency	%	*[(Total Offshore Production - Production Loss) / Total Production]*
12	Onshore Process Efficiency	%	*[(Total Onshore Production - Production Loss) / Total Production]*
13	Utility Efficiency	%	*[(Total Utility Production - Utility Production Loss) / Total Utility Production]*
14	Cumulative Production Efficiency	%	*[Cumulative Production /Planned Cumulative Production]*
15	Standby Operational Availability	%	*[(Running Time + Standby Time) /Total Time]*
16	Operational Availability	%	*[(Total Time - Scheduled Downtime - Unscheduled Downtime) / Total Time]*
17	Reliability	*%*	*[(Total Time - Unplanned Maintenance Time) / Total Time]*
18	Usage Factor	*%*	*[Running Time / Total Time]*

Tanks. Tank Levels: Shows tank levels (in %) vs time for any selected tank (storage element), identifies any time-based variability. If a tank remains most of the time either full or empty, then the tank could be a system constraint.

Warehouse Sparing. Spare Coverage: Quantifies the number of items to stock in order to have x% confidence of not having a stockout, also used as starting point for Spare Optimization.

Maintenance Costs. Maintenance Cumulative Costs: Quantifies the average repair and spare costs incurred during the lifecycle duration per event/equipment, response type (planned vs unplanned), and per unit.

Maintenance Total Costs Per Event: Shows variability in repair + spare costs per event/equipment.

Conclusions

The highly-integrated model and detailed simulation approach offers a powerful framework to understand what happens if a specific equipment fails, operating conditions or feedstock changes, and the effect of different operation/maintenance strategies at the production area level, stream level, or enterprise level.

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This approach ensures a thorough understanding of system behaviors, risk factors, and operational dynamics, leading Informed Standardized Decision-Making Process to optimize production & maintenance processes and strategy, enhancing system reliability, availability, and maintainability, ultimately achieving a resilient and efficient production environment.

Further it is intended to expand the integrated Models to all ADNOC Offshore sites including Lower Zakum- Das and SARB streams, include electrical systems forecast Maintenance spend and additional KPI.

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Appendix

Detailed description of flow diagram elements:

Units

Units used to group assets to functionally related sets of equipment. Equipment items placed inside units. The arrangement and connectivity of equipment within units captures reliability dependencies, redundancies, capacities, and current operational states of equipment. The behavior of each unit is determined by the internal states and structure of equipment inside it and by any external constraints imposed by virtue of that unit's connections to other flow diagram elements.

A particularly important rule regarding behavior of Fidelis units is that they implement linear transfer functions. More specifically, this means that units impose a strictly proportional relationship among inputs and outputs. The proportionality is based on values of nominal flow rates assigned to pipes. E.g., if a feed pipe to a unit can only deliver 50% of its nominal flow to the receiving unit, then all other inputs and outputs to/from that unit will be turned down to 50% of their nominal values. The unit will then be operating at 50% of its nominal capacity (assuming it is internally capable of operating at that rate and no external constraints are more limiting).

Pipes

Pipes used as connections among units, manifolds, and tanks are realized by pipes which are the lines visible in Fidelis flow diagram. Typically, pipes are direct analogs to physical pipes in real systems. They are generally used to model flow of bulk materials. Fidelis Pipes have these attributes: nominal flow rate, maximum flow rate, point of origin, and point of termination.

Pipes can also model flows of other things. Electric power, financial currency, and availability of other units are examples that illustrate how flexible Fidelis pipes can be in practice.

Manifolds

Manifolds serve a modeling purpose nearly identical to what real-world headers do in piping systems. One or more pipes provide flow into manifolds and one or more pipes route flow from manifolds. Conservation rules apply to manifolds: the sum of inputs must always equal the sum of outputs at every point in time. Routing of material through manifolds depends on pipe capacities, current flows of pipes connected to the manifold, states of units that supply flow to, or receive flow from, manifolds, and on flow priorities defined in the Objectives Function.

Tanks

Tanks are high fidelity representations of their physical analogs. They store bulk material. Fidelis explicitly tracks flow into and out of tanks. The inflow/outflow difference is used to determine current level of material in a tank and net flow. Net flow can be positive (tank is filling), negative (tank is emptying), or zero (task level is not changing). A tank initially characterized by parameters called usable volume, setpoint, and initial volume. Outputs from tanks are assumed to be fully mixed when exiting. When a tank reaches a full, empty, or setpoint state, an event generated that allows users to exercise custom logic if needed to react properly to the situation.

Details of description Unit diagram elements:

e) Start and End Node - Represents the source and sink of your unit diagram.

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f) Event - Represents individual pieces of equipment events, or preventative maintenance schedule or external disturbances.
g) Rate Multiplier - Represents a header to create parallel events in the unit diagram.
h) Link - Represents transportation of available capacity information from one unit element to another unit element.

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